COMPUTER SCIENCE – FORM ONE

CHAPTER FIVE: COMPUTER SYSTEM HANDLING AND CARE

Teacher Presentation Script – Pages 102–113

How to use: These are words you can use directly when presenting the topic to students with a low level of understanding. Ask questions, give the examples, then write the key point on the board.

1. STARTING THE LESSON

Teacher: **"Good morning, students."**

Teacher: **"Today we are going to learn about Computer System Handling and Care."**

Teacher: **"Before we start, think about this: If you buy a new phone or computer, what do you do to make sure it does not get damaged?"**

After answers: **"Good. We protect it, clean it, handle it carefully and avoid things that can damage it. A computer is also an electronic device, so we must take care of it."**

Teacher: **"Today we are going to learn how to take care of a computer, what can damage it, and how we can prevent that damage."**

2. MEANING OF COMPUTER HANDLING AND CARE

Teacher: **"Computer handling and care means using, handling, protecting and maintaining a computer properly so that it can work well and for a long time."**

Example: **"If you have a laptop, you should not throw it, pour water on it, hit the keyboard or use it carelessly."**

Ask: **"Why do we need to take care of a computer?"**

Conclude: **"Exactly. We take care of a computer to prevent damage and make it continue working properly."**

3. LIQUID DAMAGE

Teacher: **"What happens when water enters a computer?"**

Teacher: **"Water and other liquids can damage computer components because computers use electricity. When liquid reaches electronic components, it may cause a short circuit and damage the computer."**

Example: **"Imagine you are using a laptop and you put a glass of water beside it. You accidentally hit the glass and the water enters the keyboard. What can happen?"**

Expected: **"The computer can stop working."**

Key point: **"Never put liquids near a computer."**

4. CORROSION AND DATA LOSS

Teacher: **"Another problem caused by liquid is corrosion. Corrosion means gradual damage of metal, especially when it is exposed to moisture."**

Example: **"Have you ever seen an old iron object that has started developing rust? That is an example of corrosion."**

Teacher: **"Liquid can also damage storage devices. If the storage device is damaged, we may lose access to our files."**

Example: **"If your assignment is stored only on a damaged computer, you may lose the assignment."**

5. WHAT TO DO AFTER A LIQUID SPILL

Write on board: **OFF → UNPLUG → DISCONNECT → DRY CAREFULLY / GET HELP**

Teacher: **"If liquid spills on a computer, first turn it off."**

Teacher: **"Second, disconnect it from electricity."**

Teacher: **"Third, disconnect external devices where it is safe to do so."**

Teacher: **"For a small external spill, carefully dry the accessible surface with a soft cloth. If a large amount of liquid enters the computer, seek help from a qualified technician."**

Emphasise: **"Do not continue using a wet computer."**

6. PREVENTING LIQUID DAMAGE

Teacher: **"How can we prevent liquid damage?"**

Explain: **"Keep drinks away from computers. Work in a clean and dry place. Do not put cups or bottles beside the computer."**

Memory: **"NO DRINKS NEAR COMPUTERS."**

7. POWER MANAGEMENT

Teacher: **"A computer needs a suitable and stable supply of electricity. When electricity becomes too low or too high, it can affect the computer."**

Teacher: **"There are three important power problems: brownout, spike and surge."**

8. BROWNOUT

Teacher: **"A brownout is a reduction in electrical voltage that lasts for some time."**

Simple explanation: **"A brownout means electricity becomes weaker than normal."**

Example: **"Lights may become dim or a fan may become slower."**

Ask: **"Brownout means high voltage or low voltage?"**

Answer: **"Low voltage."**

Board: **BROWNOUT = LOW VOLTAGE**

9. SPIKE

Teacher: **"A spike is a sudden increase in voltage that lasts for a very short time."**

Example: **"Electricity goes off and then suddenly comes back. There can be a sudden voltage increase."**

Ask: **"Is a spike a sudden decrease or increase?"**

Answer: **"Increase."**

Board: **SPIKE = SUDDEN INCREASE**

10. SURGE

Teacher: **"A surge is a significant increase in voltage that can damage electronic equipment."**

Example: **"If electricity suddenly becomes much higher than normal, sensitive computer components may be damaged."**

Board: **SURGE = LARGE SUDDEN VOLTAGE INCREASE**

11. UPS

Teacher: **"How can we protect a computer when electricity goes off?"**

Teacher: **"One device we can use is called a UPS – Uninterruptible Power Supply."**

Teacher: **"A UPS provides temporary backup power when the main electricity supply fails."**

Example: **"If electricity goes off while you are typing an assignment, a UPS can keep the computer running temporarily. You can save your work and shut down safely."**

Ask: **"What is the main purpose of a UPS?"**

Answer: **"Backup power."**

Board: **UPS = BACKUP POWER**

12. SURGE PROTECTOR

Teacher: **"A surge protector helps protect electrical equipment from voltage spikes and surges."**

Example: **"If there is a sudden voltage surge, the surge protector provides an additional layer of protection for the connected equipment."**

Board: **SURGE PROTECTOR → SURGE PROTECTION**

13. VOLTAGE STABILISER

Teacher: **"A voltage stabiliser helps maintain a more stable voltage supplied to equipment when the incoming voltage changes."**

Example: **"If incoming electricity keeps changing, a suitable stabiliser helps provide a more stable output."**

Board: **STABILISER = STABLE VOLTAGE**

14. DIFFERENCE BETWEEN THE THREE

Draw: **UPS → BACKUP POWER**

Draw: **SURGE PROTECTOR → SURGE PROTECTION**

Draw: **VOLTAGE STABILISER → STABLE VOLTAGE**

Teacher: **"If I say UPS, you say backup power."**

Students: **"Backup power!"**

Teacher: **"If I say surge protector?"**

Students: **"Surge protection!"**

Teacher: **"If I say voltage stabiliser?"**

Students: **"Stable voltage!"**

15. MOVING A COMPUTER

Teacher: **"Suppose we want to move computers from one room to another. Can we just pick them up and move them?"**

Answer: **"No."**

Explain: **"First shut down the computer. Second disconnect the cables. Third carry it carefully."**

For laptops: **"Never carry a laptop by its screen. Hold it from the base."**

Example: **"Holding a laptop by the screen may damage the display or hinges."**

16. SOFTWARE CARE AND VIRUSES

Teacher: **"We have talked about hardware. Is hardware the only thing we need to protect?"**

Answer: **"No. We also need to protect software and data."**

Teacher: **"A computer virus is harmful software that can enter a computer and cause problems."**

Ask: **"Where can viruses come from?"**

Guide: **"USB devices, infected files, unsafe downloads and other sources."**

Example: **"An infected USB can spread harmful software to a school computer."**

17. ANTIVIRUS AND SOFTWARE UPDATES

Teacher: **"Antivirus software helps detect and protect the computer from viruses and other malicious software."**

Practical sequence: **USB → SCAN → CHECK → OPEN**

Teacher: **"We should also keep antivirus software and other software updated."**

Teacher: **"Updates can provide security patches, bug fixes and improvements."**

Example: **"An important Windows security update may fix a weakness in the system."**

18. PROPER SHUTDOWN

Write: **SAVE → CLOSE → LOG OUT → SHUT DOWN → WAIT**

Teacher: **"First, save your work."**

Teacher: **"Second, close your programs."**

Teacher: **"Third, log out where appropriate."**

Teacher: **"Fourth, use the Shutdown option."**

Teacher: **"Finally, wait for the computer to finish shutting down."**

Ask: **"Why should we not simply pull out the power cable?"**

Answer: **"It can cause data loss, file corruption or system problems."**

19. KEYBOARD AND MOUSE CARE

Teacher: **"Type gently on the keyboard. Do not hit the keys or put heavy objects on it."**

Example: **"Do not put your school bag on top of the keyboard."**

Teacher: **"Handle the mouse gently. Do not drop, throw or hit it."**

Example: **"Throwing a mouse onto a desk may damage its internal components."**

20. CABLE CARE

Teacher: **"When disconnecting a cable, hold the connector, not the wire."**

Teacher: **"Do not bend cables too much. Do not put heavy objects on them. Keep them away from sharp objects."**

Teacher: **"Check cables for cuts and exposed wires."**

Emphasise: **"Never use a cable that has exposed electrical wires."**

21. PASSWORD SECURITY

Teacher: **"A password is like a key. It controls who can access an account."**

Teacher: **"A strong password should use uppercase letters, lowercase letters, numbers and special characters. It should also be sufficiently long."**

Example: **"School@2026#Lab is only a classroom example. Do not use classroom examples as your real password."**

Teacher: **"Avoid simple passwords such as 123456, password, your name or birthday."**

22. UNIQUE PASSWORDS AND 2FA

Teacher: **"Do not use the same password for every important account."**

Example: **"If the same password is used for several accounts and it becomes known, several accounts may be at risk."**

Teacher: **"Two-factor authentication, or 2FA, adds another verification step after the password."**

Example: **"Password → verification code → access."**

Memory: **"Password = first lock. 2FA = second lock."**

23. PHISHING

Teacher: **"Phishing is an attempt to trick someone into giving away sensitive information such as a password."**

Example: **"A message says: Your account will close today. Click this link and enter your password. Should you immediately enter it?"**

Expected: **"No."**

Teacher: **"Correct. Verify the message and website before entering sensitive information."**

24. BACKUP

Teacher: **"What happens if the computer suddenly stops working and all your files are stored only on that computer?"**

Expected: **"We can lose the files."**

Teacher: **"Exactly. A backup is an additional copy of important data."**

Example: **"Keep another copy of an important assignment so that it can be recovered if the computer fails."**

Board: **BACKUP = EXTRA COPY**

25. PHYSICAL SECURITY

Teacher: **"Physical security means protecting computer equipment from theft, unauthorised physical access and physical damage."**

Example: **"A school computer laboratory can be locked when it is not in use."**

Teacher: **"A cable lock can attach a laptop to a fixed object such as a desk."**

Teacher: **"Other physical security measures include locked doors, CCTV, alarms and access control."**

Key point: **"Computer security is not only about passwords. We must also protect the physical environment."**

26. FINAL REVISION

Teacher: **"Let us see whether you have understood the lesson."**

1. **"What is a brownout?"** — Low voltage.
2. **"What is a spike?"** — A sudden increase in voltage.
3. **"What is the purpose of a UPS?"** — Backup power.
4. **"What does a surge protector do?"** — Helps protect against surges and spikes.
5. **"What does a voltage stabiliser do?"** — Helps maintain stable voltage.
6. **"Why do we use antivirus?"** — To help protect against viruses and malware.
7. **"Why should we shut down properly?"** — To reduce data loss, file corruption and system problems.
8. **"What is 2FA?"** — Two-factor authentication.

9. **"What is a backup?"** — An additional copy of data.

10. **"What is physical security?"** — Protecting equipment from theft, unauthorised physical access and physical damage.

27. CLOSING THE LESSON

Teacher: **"Students, today we have learned that a computer is an electronic device and therefore it needs proper care."**

Teacher: **"We have learned that liquid can damage computer components and cause data loss."**

Teacher: **"We have learned about brownouts, spikes and surges, and how UPSs, surge protectors and voltage stabilisers help with power problems."**

Teacher: **"We have also learned how to handle computers, keyboards, mice and cables carefully."**

Teacher: **"We have learned how to protect software using antivirus and updates, and how to protect information using strong passwords, 2FA and backups."**

Teacher: **"Finally, we have learned about physical security and protecting computers from theft and unauthorised access."**

Final message: **"Protect the computer, protect the software, protect the data, and protect the user."**

BOARD SUMMARY

COMPUTER CARE

Hardware → Liquid, Power, Keyboard, Mouse, Cables

Power → Brownout, Spike, Surge

UPS → Backup power

Surge protector → Surge protection

Stabiliser → Stable voltage

Software → Virus, Antivirus, Updates

Safe use → Save, Close, Log out, Shut down, Wait

Account security → Strong password, Unique password, 2FA, Phishing awareness

Data → Backup

Physical security → Locks, CCTV, Alarms, Access control